

# L9: Number of Partners in Sexual Networks

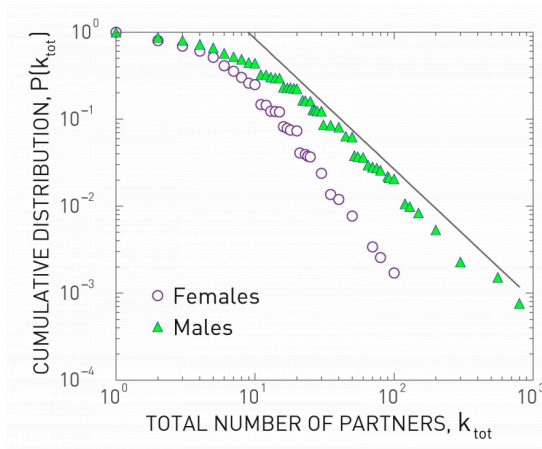


Figure 10.13 from Network Science by Albert-László Barabási

All previous derivations assume that each individual has the same number of contacts. This assumption makes the derivations simpler – but as we will see later in this lesson, it can also be misleading especially when the number of contacts of different individuals varies considerably.

Let us start with sexually transmitted diseases. The plot at the left shows the distribution of the number of sexual partners, separately for men and women, since sexual initiation in a 1996 survey in Sweden. Note that the plot is in log-log scale. The straight-line decay when the number of partners is larger than 20 suggests that the corresponding distribution follows a power-law. The exponent is about 3 for women and 2.6 for men.

Even though most men had less than 10-20 partners, there are also individuals with 100s of partners.

(To the right: Figure 10.14 from Network Science by Albert-László Barabási.)

The plot at the right shows is based on a survey of high school students and “romantic relationships”. Note that even though there are 63 couples without any other connections and many other nodes with only 1-2 connections, there are also few nodes with a much higher number of such relationships (*up to almost 10*).

Assuming that every individual has the same number of contacts/partners would be very far from the truth at least in these two datasets. So, clearly, the homogeneous mixing assumption is very unrealistic.

